

energy being tapped in large areas other than the terrace! Believed to be one of the largest installations of its kind, it is hoped that the captured energy will provide for 15 per cent of the campus' power needs. During very sunny periods, it might even provide 50 per cent of the required power. It is believed that the investment will breakeven in as early as three years.

4 Hot or cold, keep it within your control

Notable energy-guzzlers, be it at your home or in a commercial establishment, are the HVAC systems. As a natural consequence of the ongoing green drive, various manufacturers are coming out with energy-efficient HVAC systems.

For some time now, energy-saving features have been there in air-conditioners (ACs): sleep-mode timers, temperature control, etc. Of late, ACs are also being fitted with so-called 'magic eyes' or motion sensors, which detect the presence of people and lower the temperature setting or capacity automatically when the occupants leave the room, reverting to the presets as soon as the room is occupied again.

Brushless DC motors for air-conditioning compressors and built-in inverter control are the other energy-saving technologies taking over the air-conditioning space. Though these technologies have been spoken about since 2005, it's only now that they have become mainstream. Inverter-fitted ACs, as against older models, can work in variable capacities. So once the required temperature is attained, these ACs simply shift to a lower capacity rather than shutting down and restarting the compressor—a process that consumes a lot more energy.

Industrial HVAC systems implement such mechanisms on a more complex scale, having a whole network of sensors to monitor all areas, centrally control (or even automate) temperature settings and so on. Nowadays, energy management systems can even be accessed over the Web from remote locations, to monitor

Lifeline technologies for a green building

Some of the most exciting technologies like sensor networks, embedded electronics, embedded algorithms and software are maturing to become the lifeline for green buildings.

Here is a sample of how these act as enablers:

1. Solar energy: Improve solar cell (conversion) efficiency
2. Smart grids: Automatic metering infrastructure (AMI), alternative power generation, usage profiling and consumption control
3. Quality of occupancy: Ambient intelligence, lighting (LED, powerline communications), temperature control, security
4. Water consumption: Water metering, quality assessment
5. Micro green zones: Increasing the green canopy in enclosed spaces
6. Embedded software: Algorithms, network creation, maintenance and analytics

—S. Uma Mahesh, co-founder and CEO of Indrion Technologies (provider for green buildings)

or vary energy consumption. Computerised energy audits also help to take a critical view of energy consumption and redesign systems or recalibrate settings accordingly.

5 Building-integrated and building-wide green technologies

There are several building-integrated 'green' technologies that are available today: high albedo paints and other materials to keep buildings cool, automatic and tight doors and windows that can avoid energy leakage, better design of rooms to provide natural light and ventilation, lighting systems, intelligent wiring, rainwater harvesting, green roofs, solar panels and so on. Your budget, of course, might shoot up anywhere from 3 to 10 per cent, but the investment is worth it because the savings in the long run will be much more!

If you are also considering futuristic technologies, solar-panelled windows are a good option. Konarka has developed flexible, transparent plastic-based solar cells that can be used as a layer on windows. This technology is soon going to be used by a Massachusetts-based company to manufacture building materials, including windows.

Thinking ahead of windows, what if there is a paint that can harness solar energy? Several teams of researchers (including one from Swansea University and the other from New Jersey Institute of Technology) have demonstrated the possibility.

Building-integrated technologies

apart, there are several building-wide technologies that one could adopt: lighting systems, HVAC control systems, sensor networks, other energy management solutions, smart meters that help to analyse and understand power consumption, optimised wiring design and so on.

On governments and local markets

With abundant natural resources and new constructions happening all the way, we are well set to embark on a revolution. However, we need a push. Where could it come from, and how did other countries, such as Germany, manage to accelerate the trend?

"Green technologies will become affordable on the hinge of two requirements: first, it is essential that governments commit to green technology within their jurisdictions, and second, local markets and industries must be developed which can support regional economic and sustainable growth and development," answers Christine Thuring, contributing editor to Greenroofs.com and director of Chlorophyllocity.

"India has one of the highest GDPs in the world. Consider the exponential increase of green buildings in the last few years, and the Indian Green Building Council's activity since its inception. If these conditions continue, it may be plausible that India could have an all-green construction system in 60 years' time." ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Bengaluru